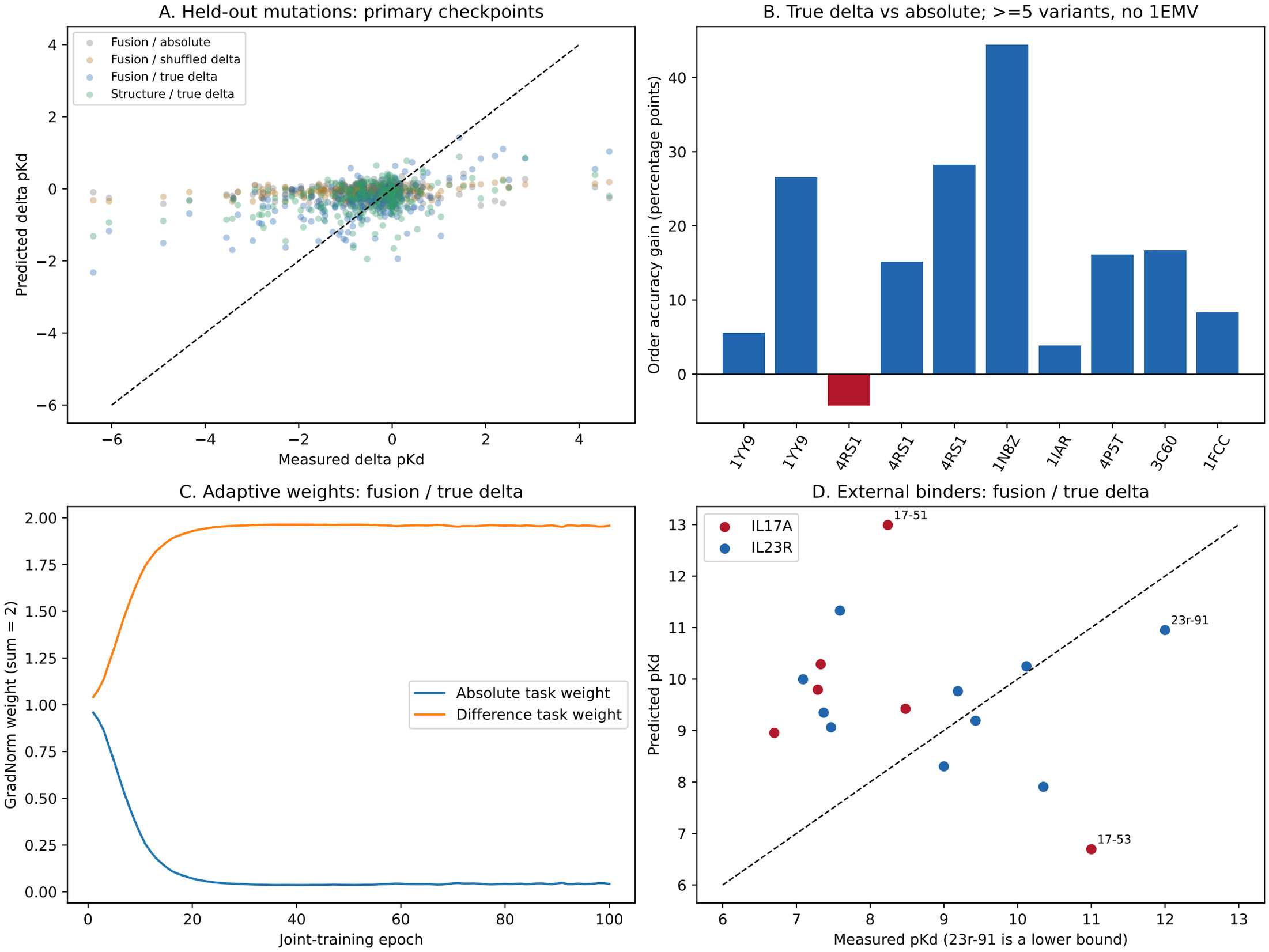
A. Held-out mutations: primary checkpoints
Fusion / absolute
Fusion / shuffled delta
Fusion / true delta
Structure / true delta
Predicted delta pKd
Measured delta pKd
B. True delta vs absolute; >=5 variants, no 1EMV
Order accuracy gain (percentage points)
1YY9
1YY9
4RS1
4RS1
4RS1
1N8Z
1IAR
4P5T
3C60
1FCC
C. Adaptive weights: fusion / true delta
GradNorm weight (sum = 2)
Absolute task weight
Difference task weight
Joint-training epoch
D. External binders: fusion / true delta
IL17A
IL23R
17-51
23r-91
17-53
Predicted pKd
Measured pKd (23r-91 is a lower bound)